

PHYS 230 - Electricity and Magnetism

Lecture 9

Instructor: Saeed Rastgoo



**UNIVERSITY
OF ALBERTA**

Outline

2 Gauss's Law

- Electric Flux (Continued)
- Gauss's Law

All figures in these slides are from one of the following sources, unless cited otherwise:

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Physics for Scientists and Engineers, Hawkes, et al.

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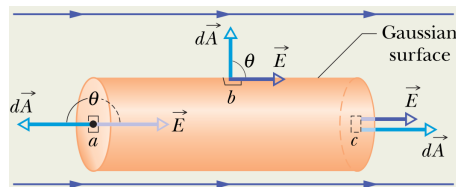
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Exercise 1

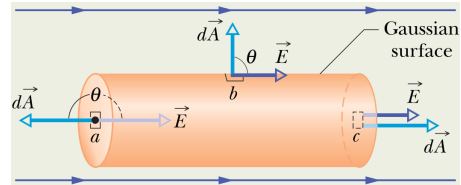
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Figure shows a closed surface in the form of a cylinder of radius R and height H immersed in a uniform electric field $\vec{E}$, with the cylinder axis parallel to the field. What is the flux of the electric field through this closed surface?



Exercise I

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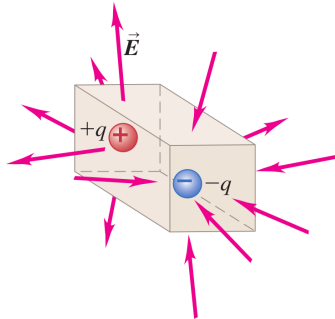
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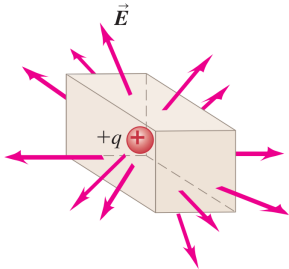
Zero *net* charge inside box,
inward flux cancels outward flux.



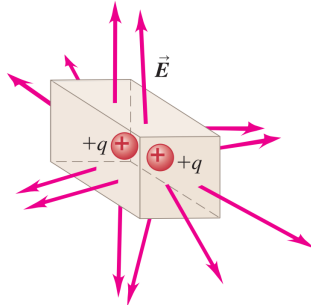
Flux of Closed Surfaces With No Net Charge in Them

- From the examples: **Electric field flux through closed surfaces enclosing no net charge is zero!**
- This is actually always true: If a field line enters the volume enclosed by the surface, it will exit from some other parts of the surface $\Rightarrow$ no net flux
- If there is a net charge q inside, then more lines either enter or exit $\Rightarrow$ net non-zero flux

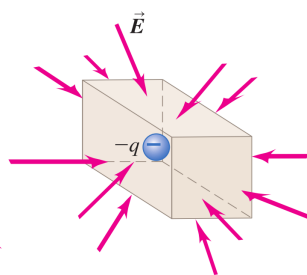
(a) Positive charge inside box, outward flux



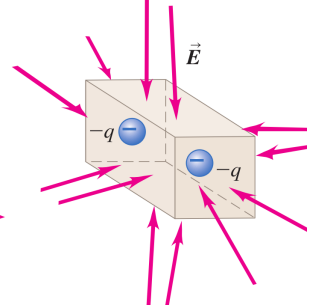
(b) Positive charges inside box, outward flux



(c) Negative charge inside box, inward flux



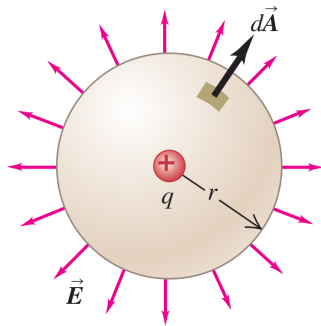
(d) Negative charges inside box, inward flux



Exercise 2

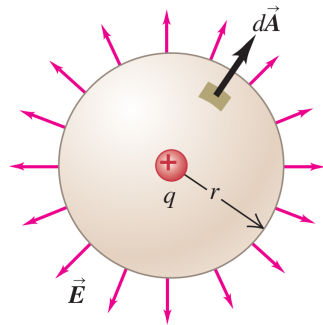
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A point charge $q = +3.0 \mu\text{ C}$ is surrounded by an imaginary sphere of radius $r = 0.20\text{ m}$ centered on the charge (see figure). Find the resulting electric flux through the sphere.



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Gauss's Law

The total electric flux through any closed surface (a surface enclosing a definite volume) is equal to the total (i.e., net) electric charge inside the surface, divided by ϵ_0

$$\underbrace{\oint \mathbf{E} \cdot d\mathbf{A}}_{\Phi} = \frac{q_{\text{in}}}{\epsilon_0}$$

where $\oint$ means integral over a closed surface, and $q_{\text{in}} = q_1 + \dots + q_n$ is the net charge inside the volume enclosed by the surface

Gauss's Law

Equivalent forms of Gauss's Law

$$\oint \mathbf{E} \cdot d\mathbf{A} = \frac{q_{\text{in}}}{\epsilon_0}$$

or

$$\oint \mathbf{E} \cdot \mathbf{n} dA = \frac{q_{\text{in}}}{\epsilon_0}$$

or

$$\oint E \cos(\theta_{\mathbf{E}, d\mathbf{A}}) dA = \frac{q_{\text{in}}}{\epsilon_0}$$

Gauss's Law

Gauss's Law: an example of bulk-boundary correspondence!

$$\underbrace{\oint \mathbf{E} \cdot d\mathbf{A}}_{\text{info on boundary}} = \frac{\overbrace{q_{\text{in}}}^{\text{info about bulk}}}{\epsilon_0}$$

Gauss's Law

In general $\mathbf{E}$ depends on the point on the surface we are considering (i.e., $\mathbf{E}$ is not the same or constant all over the surface)

$\Rightarrow$ cannot bring $\mathbf{E}$ out of the integral

$$\oint \mathbf{E} \cdot d\mathbf{A} = \frac{q_{\text{in}}}{\epsilon_0}$$

Gauss's Law

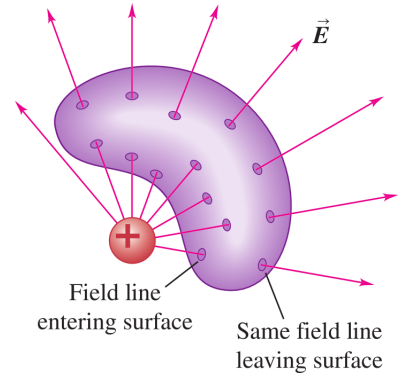
Example: if the surface does not contain any (net) charge

$$\oint \mathbf{E} \cdot d\mathbf{A} = 0$$

or

$$\Phi = 0$$

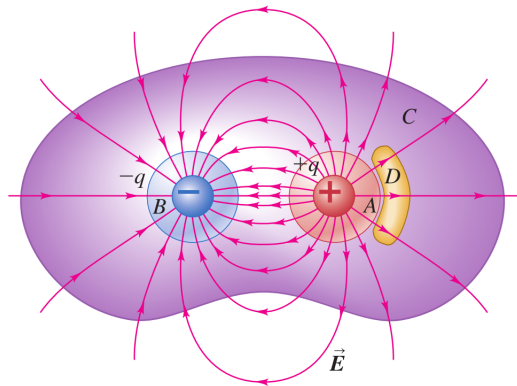
Figure 22.13 A point charge *outside* a closed surface that encloses no charge. If an electric field line from the external charge enters the surface at one point, it must leave at another.



Exercise 3

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Find the electric flux through each of the closed surfaces A, B, C, and D.



Applying Gauss's Law

Gauss's law

$$\underbrace{\oint \mathbf{E} \cdot d\mathbf{A}}_{\Phi} = \frac{q_{\text{in}}}{\epsilon_0}$$

can be used in three ways:

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1. If q_{in} is known, we can directly find the flux $\Phi = \frac{q_{\text{in}}}{\epsilon_0}$

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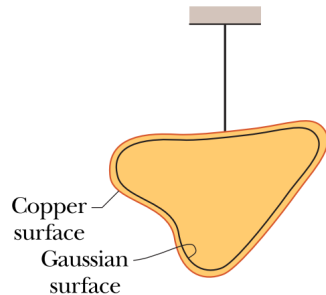
can be used in three ways:

1. If q_{in} is known, we can directly find the flux $\Phi = \frac{q_{\text{in}}}{\epsilon_0}$
2. If q_{in} is known and there is enough symmetry, we can find $\mathbf{E}$
3. If $\mathbf{E}$ is known, we can integrate and find q_{in}

Exercise 4

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Show that if an excess charge is placed on an isolated conductor, that amount of charge will move entirely to the surface of the conductor. None of the excess charge will be found within the body of the conductor.



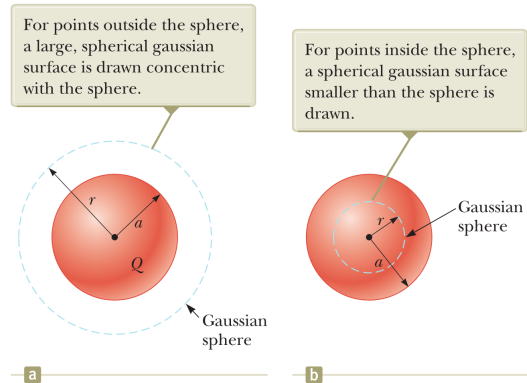
Exercise 5

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An insulating solid sphere of radius a has a uniform volume charge density ρ and carries a total positive charge Q (see figure).

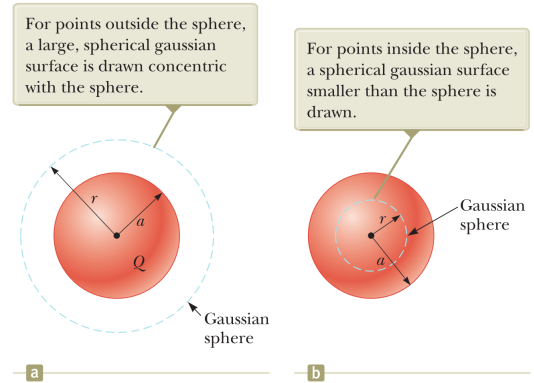
(a) Calculate the magnitude of the electric field at a point outside the sphere.

(b) Find the magnitude of the electric field at a point inside the sphere.



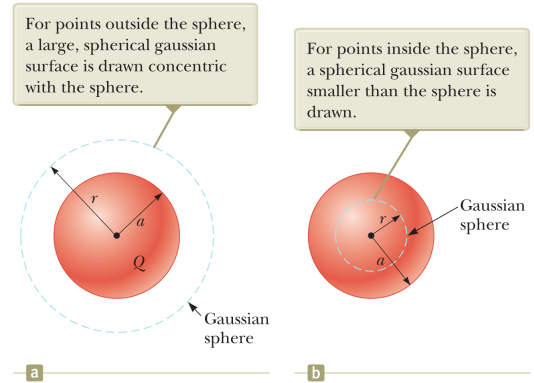
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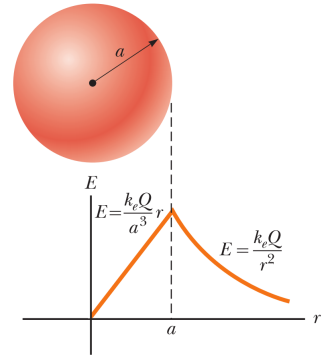


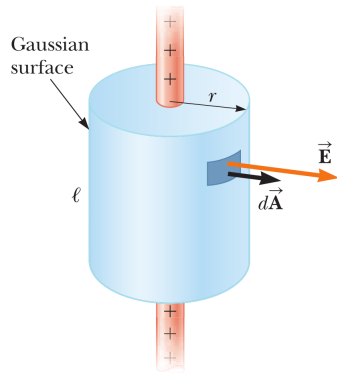
Figure 24.11 (Example 24.3)

A plot of E versus r for a uniformly charged insulating sphere. The electric field inside the sphere ($r < a$) varies linearly with r . The field outside the sphere ($r > a$) is the same as that of a point charge Q located at $r = 0$.

Exercise 6

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Find the electric field at a distance r from a line of positive charge of infinite length ℓ and a constant charge per unit length λ .



Exercise 6

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